

Model Selection and Cross-Validation

Alexander McLain

May 29, 2026

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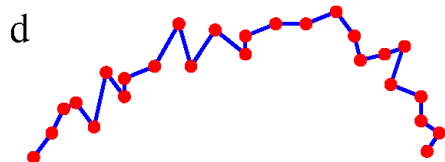
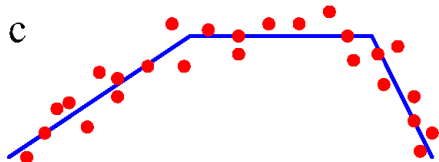
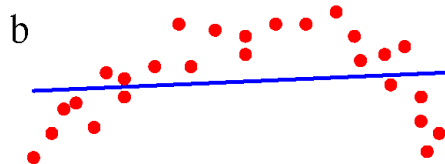
The Problem

- ▶ Yesterday we fit regression models with multiple predictors.
- ▶ In practice, you often have **many** candidate predictors and need to decide which ones to include.
- ▶ **Model selection** = choosing the right subset of variables for your model.
- ▶ Why it matters:
 - ▶ Too few predictors $\Rightarrow$ missing important patterns (**underfitting**)
 - ▶ Too many predictors $\Rightarrow$ fitting noise instead of signal (**overfitting**)
- ▶ Goal: find the balance between simplicity and accuracy.

Example: Predicting Blood Pressure

- ▶ You want to predict systolic blood pressure.
- ▶ Available predictors: age, weight, BMI, cholesterol, cigarettes/day, glucose, resting pulse, education, sex, diabetes status. . .
- ▶ Which should you include in the model?
- ▶ Including everything is tempting, but:
 - ▶ Some predictors add noise, not signal.
 - ▶ Correlated predictors inflate uncertainty.
 - ▶ More variables = harder to interpret.
- ▶ We need **principled criteria** for choosing.

Underfitting vs. Overfitting



Underfitting vs. Overfitting: What Goes Wrong

- ▶ **Underfitting** = important predictors are left out of the model.
 - ▶ The model misses real patterns in the data.
 - ▶ Predictions are systematically biased.
 - ▶ Example: predicting BP from age alone when BMI and smoking also matter.
- ▶ **Overfitting** = the model includes predictors that capture noise, not signal.
 - ▶ The model fits the *training data* very well but predicts *new data* poorly.
 - ▶ Coefficients become unstable and hard to interpret.
 - ▶ Confidence and prediction intervals become unnecessarily wide.

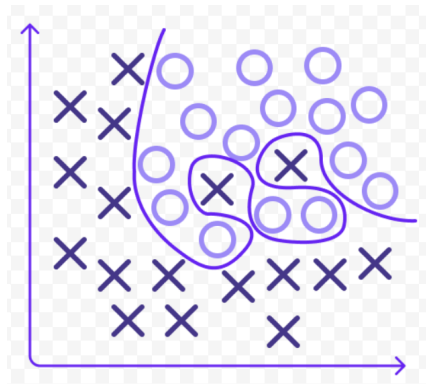
Memorization vs. Generalization

Memorization (Overfitting)

- ▶ Fits the training data perfectly.
- ▶ Captures noise and quirks of the sample.
- ▶ Fails on new data.

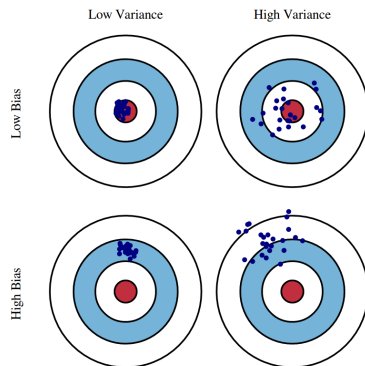
Generalization (What We Want)

- ▶ Captures the *real structure* in the data.
- ▶ Training error is low *and stays low* on new data.
- ▶ This is the goal of model selection.



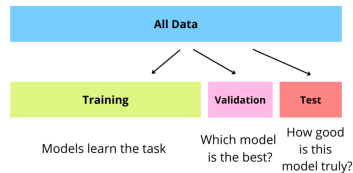
The Bias–Variance Tradeoff

- ▶ **Bias:** systematic error from a model that is too simple. The model consistently misses the target.
- ▶ **Variance:** instability from a model that is too complex. The model is sensitive to small changes in the data.
- ▶ Adding complexity **reduces bias** but **increases variance**.
- ▶ The **sweet spot** minimizes the total prediction error.
- ▶ **Key insight:** the best model is not always the most complex one — it is the one that generalizes best.



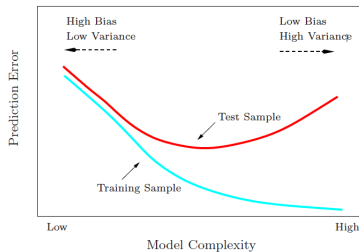
Training Data vs. Test Data

- ▶ **Training data:** the observations used to fit the model (estimate $\hat{\beta}$).
- ▶ **Test data:** observations the model has *never seen* — used only to evaluate prediction accuracy.
- ▶ A model always looks better on the data it was trained on than on new data. This is normal.
- ▶ The question is: **how much worse** does it get on new data?
- ▶ If the gap is small $\Rightarrow$ the model generalizes well.
- ▶ If the gap is large $\Rightarrow$ the model is overfitting.



Metric	Meaning
Training error	How well the model fits the data it was trained on
Test error	How well the model predicts data it has never seen

Training Error vs. Test Error



- ▶ Training error always decreases as you add predictors.
- ▶ Test error (on new data) decreases at first, then **increases** as the model starts fitting noise.
- ▶ The gap between training and test error is the signature of overfitting.

Two Approaches to Model Selection

- ▶ **Approach 1: Information Criteria (AIC, BIC)**
 - ▶ Use a formula to penalize model complexity.
 - ▶ Fast — no need to refit models on subsets of data.
 - ▶ Good for comparing a moderate number of candidate models.
- ▶ **Approach 2: Cross-Validation**
 - ▶ Directly estimate how well the model predicts new data.
 - ▶ More computationally intensive.
 - ▶ More general — works for any type of model.
- ▶ In practice, both are widely used. Today we cover both.

The Idea Behind Information Criteria

- ▶ R^2 always increases when you add a predictor — even a useless one. So R^2 alone cannot tell you the best model.
- ▶ Information criteria solve this by adding a **penalty** for the number of parameters in the model.
- ▶ The penalty discourages overfitting: a predictor must improve the fit *enough* to justify its inclusion.
- ▶ **Lower values are better** for both AIC and BIC.

AIC and BIC

- ▶ **AIC** (Akaike Information Criterion):

$$AIC = -2 \log(\text{likelihood}) + 2p$$

where p is the number of estimated parameters.

- ▶ **BIC** (Bayesian Information Criterion):

$$BIC = -2 \log(\text{likelihood}) + p \log(n)$$

where n is the sample size.

- ▶ BIC penalizes complexity **more heavily** than AIC (because $\log(n) > 2$ for $n \geq 8$).
- ▶ In practice:
 - ▶ AIC tends to select slightly larger models.
 - ▶ BIC tends to select simpler models.
 - ▶ Both are widely used; neither is “right” in all cases.
- ▶ In R: `AIC(model1, model2)` and `BIC(model1, model2)`.

Interpreting AIC/BIC Differences

- ▶ The *absolute* values of AIC and BIC are meaningless — only *differences* between models matter.
- ▶ Rough guidelines for AIC differences (ΔAIC):
 - ▶ $\Delta < 2$: Models are essentially equivalent.
 - ▶ $2 < \Delta < 10$: Some evidence for the lower-AIC model.
 - ▶ $\Delta > 10$: Strong evidence for the lower-AIC model.
- ▶ When two models have similar AIC, prefer the simpler one (principle of parsimony).
- ▶ AIC and BIC can only compare models fit to the **same data** — never compare across different datasets or different outcome variables.

Using AIC for Stepwise Selection in R

- ▶ R's `step()` function automates model selection using AIC.
- ▶ It starts with a model and iteratively adds or removes predictors, choosing the step that most improves (lowers) AIC.
- ▶ Three directions:
 - ▶ `direction = "forward"`: start empty, add predictors.
 - ▶ `direction = "backward"`: start full, remove predictors.
 - ▶ `direction = "both"`: add or remove at each step (recommended).
- ▶ Example:
 - ▶ `step(model_full, direction = "both", trace = 0)`
- ▶ `trace = 0` suppresses the step-by-step output; set `trace = 1` to watch the algorithm work.

Automated Model Selection: The Idea

- ▶ When you have many candidate predictors, manually comparing all possible models is impractical.
 - ▶ With $p = 10$ predictors, there are $2^{10} = 1024$ possible models.
 - ▶ With $p = 20$, there are over 1 million.
- ▶ **Stepwise selection** is a greedy algorithm that searches through models by adding or removing one predictor at a time.
- ▶ At each step, it picks the change that most improves the selection criterion (AIC or p-value).

Forward, Backward, and Stepwise Selection

► **Forward selection:**

1. Start with no predictors (intercept only).
2. Add the predictor that improves AIC the most.
3. Repeat until no addition improves AIC.

► **Backward selection:**

1. Start with all predictors in the model.
2. Remove the predictor whose removal improves AIC the most.
3. Repeat until no removal improves AIC.

► **Stepwise (both):**

1. At each step, consider both adding and removing a predictor.
2. Take whichever single change improves AIC the most.
3. Stop when no change improves AIC.

Limitations of Stepwise Selection

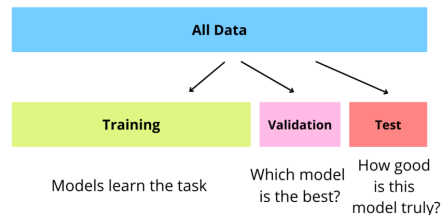
- ▶ Stepwise selection is **useful but imperfect**. Keep these limitations in mind:
 - ▶ It does not guarantee finding the globally best model — it is a greedy search that can get stuck.
 - ▶ The order in which variables enter or leave can affect the final model.
 - ▶ p-values from the final model are optimistic (biased low) because the model was chosen from many candidates.
 - ▶ Correlated predictors can cause instability — the algorithm may swap one for another across different samples.
- ▶ **Best practice:** use stepwise as a starting point, then validate with domain knowledge, diagnostics, and (when possible) cross-validation.

Why Cross-Validation?

- ▶ AIC and BIC estimate prediction error using a *mathematical formula*.
- ▶ Cross-validation estimates prediction error by *actually predicting new data*.
- ▶ The idea: if we had unlimited data, we would fit the model on one dataset and test it on a completely separate dataset.
- ▶ Since we don't have unlimited data, cross-validation **simulates** this by rotating which part of the data is used for fitting vs. testing.

Training Set vs. Test Set

- ▶ **Training error** = how well the model fits the data it was trained on (always optimistic).
- ▶ **Test error** = how well the model predicts data it was *not* trained on (what we actually care about).
- ▶ The simplest approach: split the data once (e.g., 80% train, 20% test).
- ▶ Problem: results depend on which observations end up in which group.



K-Fold Cross-Validation

- ▶ Split the data into K equally-sized groups (**folds**).
- ▶ For each fold:
 1. Hold out that fold as the test set.
 2. Train the model on the remaining $K - 1$ folds.
 3. Predict the held-out fold and record the error.
- ▶ Average the errors across all K folds.
- ▶ Common choices: $K = 5$ or $K = 10$.
- ▶ Every observation is used for testing exactly once.



What Metric to Use?

- ▶ For linear regression, the most common metric is the **mean squared error (MSE)**:

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- ▶ Often reported as **RMSE** (root mean squared error): $\sqrt{MSE}$, which is in the same units as Y .
- ▶ You can also use **MAE** (mean absolute error): $\frac{1}{n} \sum |Y_i - \hat{Y}_i|$ — less sensitive to outliers.
- ▶ The model with the **lowest CV error** is preferred.
- ▶ When two models have similar CV error, prefer the simpler one.

When to Use AIC vs. Cross-Validation

- ▶ **Use AIC/BIC when:**
 - ▶ You are comparing a moderate number of regression models.
 - ▶ You want a quick answer.
 - ▶ Your sample size is moderate to large.
- ▶ **Use cross-validation when:**
 - ▶ You want a direct estimate of prediction error.
 - ▶ You are comparing very different types of models (e.g., linear vs. spline vs. tree).
 - ▶ Your dataset is small and every observation matters.
- ▶ In many practical situations, AIC and CV will agree on which model is best. When they disagree, cross-validation is generally more trustworthy.

Common Mistakes in Model Selection

- ▶ **Using R^2 to choose a model.** R^2 always increases when you add a predictor, even a useless one. Use AIC, BIC, adjusted R^2 , or CV instead.
- ▶ **Data leakage.** If information from the test set influences how you build the model, your error estimate will be too optimistic. Example: computing the mean of the full dataset before splitting.
- ▶ **Reporting training error as performance.** The training error is always better than the true prediction error. Always report test or CV error.
- ▶ **Trusting p-values after stepwise selection.** The p-values in the final stepwise model are biased because the model was chosen from many candidates. Use them cautiously.
- ▶ **Overfitting to the validation procedure.** Trying hundreds of models and reporting the best CV error can itself overfit. Keep the number of candidates reasonable.

Google Flu Trends (2008–2014)

- ▶ **Idea:** Predict U.S. influenza levels by mining billions of Google search queries.
- ▶ **Early success:** Model looked excellent on historical data.
- ▶ **What went wrong:** The model memorized ephemeral search patterns (media hype, unrelated “flu” terms) that did not generalize. It over-predicted the 2013 flu season by $\sim 140\%$.
- ▶ **Lesson:** A model that fits past data well is not guaranteed to predict the future. Continuous validation is essential.

COVID-19 Chest X-ray Classifiers (2020)

- ▶ **Hype:** Deep learning models reported $>95\%$ accuracy distinguishing COVID pneumonia from normal lungs.
- ▶ **The problem:** Training datasets mixed adult COVID images with pediatric controls. The models learned to distinguish adults from children, not COVID from healthy.
- ▶ **Reality:** On properly age-matched test sets, accuracy dropped to near chance.
- ▶ **Lesson:** Spurious correlations hide in poorly designed datasets. Always check what your model is actually learning.

Epic's Sepsis Prediction Model (2017–2021)

- ▶ **Advertised:** $AUC \approx 0.80$ on internal validation.
- ▶ **Independent audit:** Three Midwestern hospitals found sensitivity $<40\%$ — the model missed most true sepsis cases.
- ▶ **Contributing factors:** The model was trained on one population and applied to another; internal validation was not representative.
- ▶ **Lesson:** Internal validation (even with cross-validation) does not guarantee performance in a new setting. External validation on independent data is the gold standard.

Key Themes from These Examples

- ▶ Models that fit training data well can fail catastrophically on new data.
- ▶ Spurious patterns and confounders are common, especially in large datasets.
- ▶ Internal validation is necessary but not sufficient — external validation is the gold standard.
- ▶ **The scientific question should always drive the analysis.** A statistically selected model that makes no biological sense is not useful.

Today's Key Takeaways

1. **Model selection** is about balancing fit and complexity. Neither the simplest nor the most complex model is necessarily best.
2. **AIC and BIC** penalize complexity and allow quick model comparisons. Lower is better; only differences matter.
3. **Stepwise selection** (`step()` in R) automates the search using AIC. It is useful but has limitations.
4. **Cross-validation** directly estimates how well a model predicts new data. 5-fold or 10-fold CV is standard.
5. **Training error is always optimistic.** Never use it as your measure of model performance.
6. Always combine statistical criteria with **scientific judgment** — does the model make biological sense?